Flow Distribution in Commercial Depth Filter Capsules Performance and Scale-up

Negin Nejati, Minyoung Kim, Andrew Zydney

Lara Fernandez Cerezo, Ehsan Espah, David Roush

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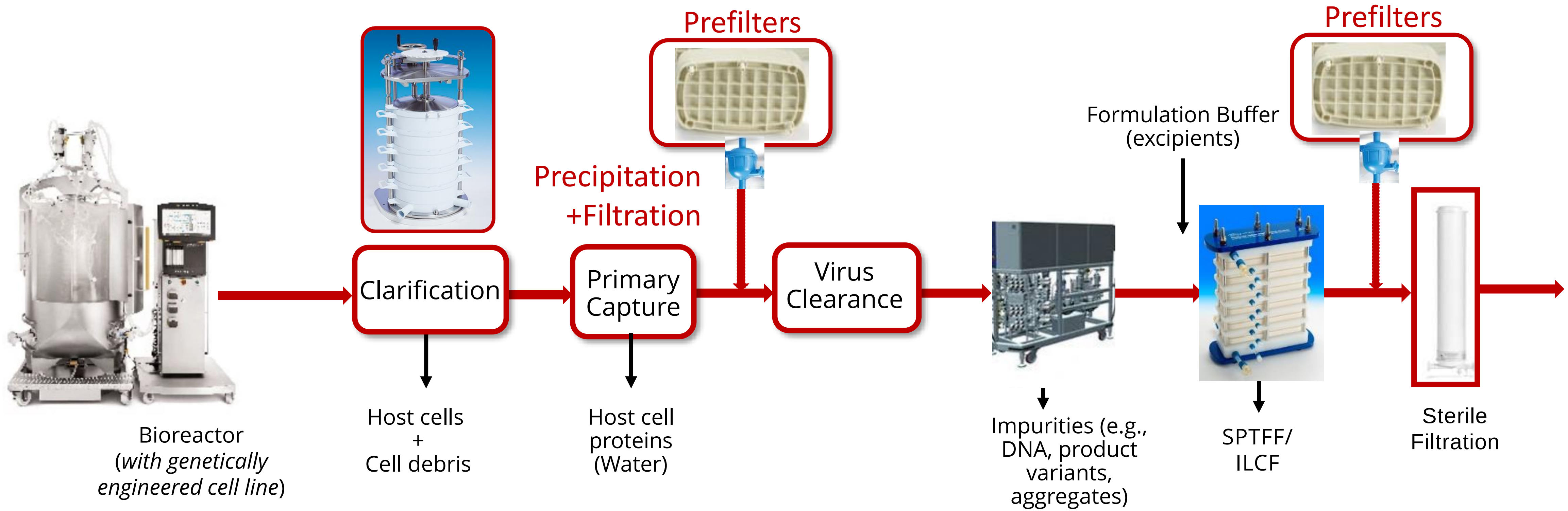
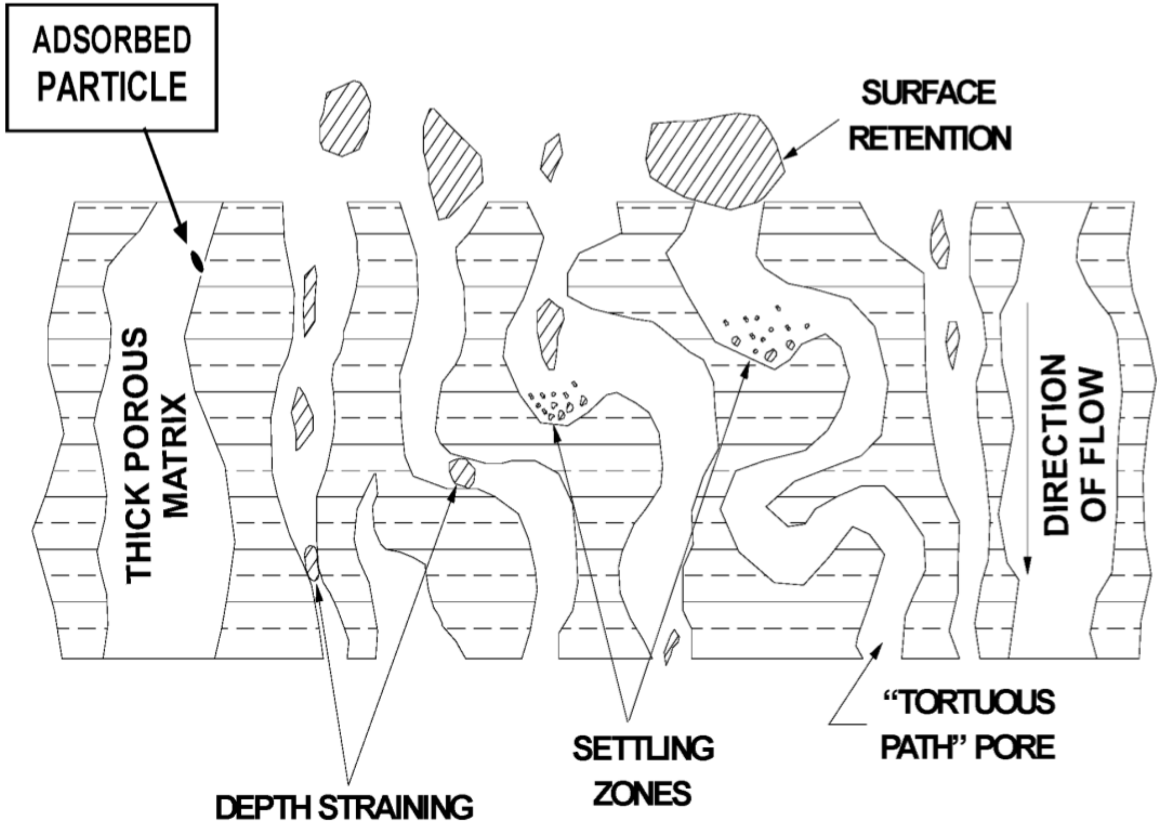
Depth Filtration in Monoclonal Antibody Processing

Depth filters used for both clarification and prefiltration



The global depth filtration market is projected to reach $2.9 Billion by 2025

Depth Filter Operations



Roush DJ, Lu Y. 2008. Advances in Primary Recovery: Centrifugation and Membrane Technology. *Biotechnol. Prog.* **24**:488–495. http://doi.wiley.com/10.1021/bp070414x.

- Species removed by a variety of mechanisms including
  - Surface retention
  - Straining (within filter)
  - Adsorption
  - Settling

- Currently: Empirical sizing with safety factors for Bulk Drug Substance

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Depth Filter Scale-Up

Clarification of 2000L bioreactor using 0.25 m^2 depth filter

- ➢ Using the manufacture recommendation and safety factor of 1.5
- ➢ **225% excess footprint!**

Supracap™ 50
Small-scale

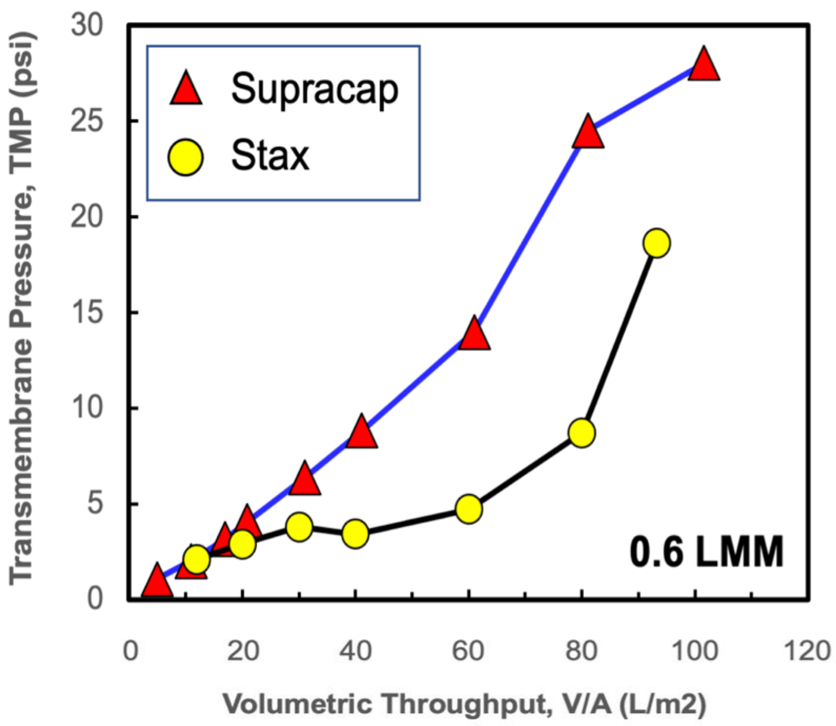
Scale-up

Stax™
Large-scale

Surface area per capsule:
$0.22 \times 10^{-4}\ \mathrm{m}^2$

Surface area per capsule:
$0.25\ \mathrm{m}^2$

TMP Profiles



Objectives

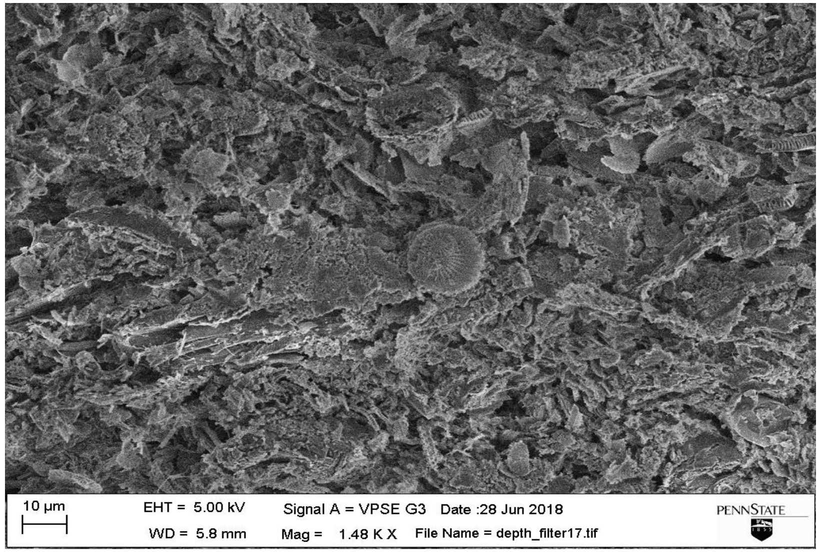
Develop a fundamental understanding of factors governing performance and scale-up of depth filtration processes using Pall PDH4 lenticular media as model system

1. Develop CFD model describing flow profiles in scale-down and pilot-scale depth filter modules
2. Validate CFD model using dye-binding and residence time distribution measurements
3. Evaluate DNA removal characteristics in both scale-down and pilot-scale modules

Model Depth Filter – PDH4 Supracap™ 50

PDH4 Supracap™ 50

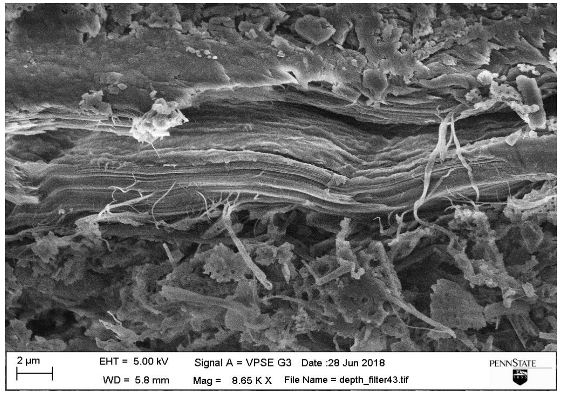
4-Layers of depth filter

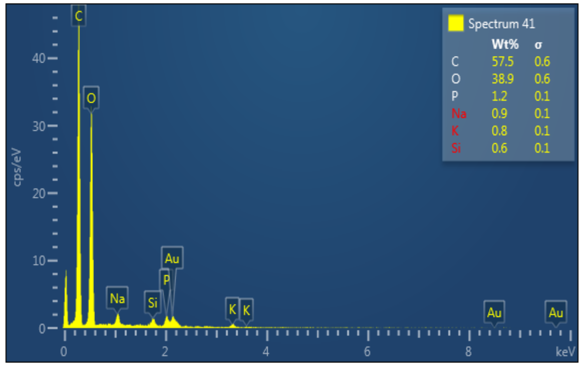


2 mm

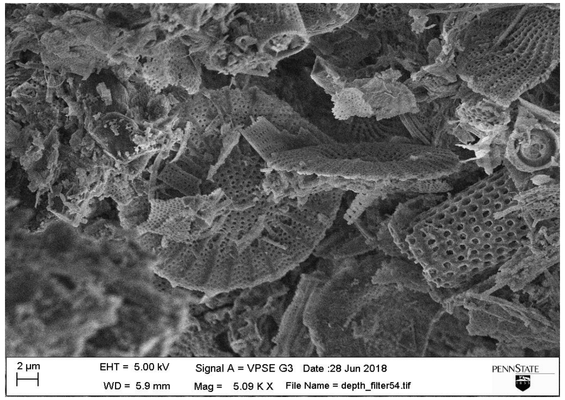
SEM image of the media cross section

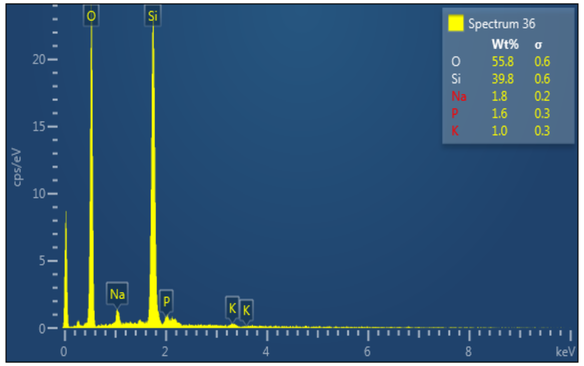
Cellulose Fibers



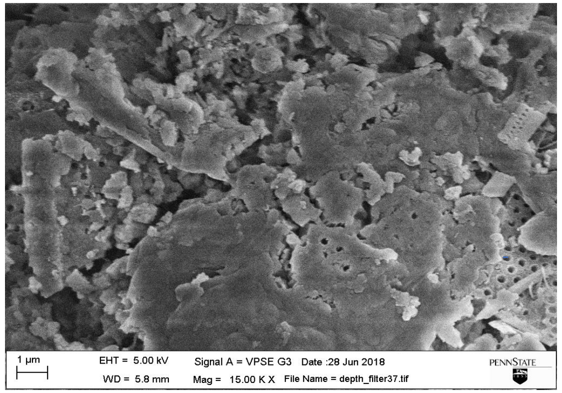


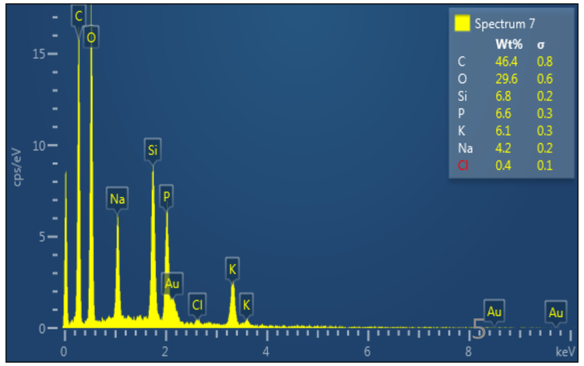
Diatomaceous Earth (DE)





A binder





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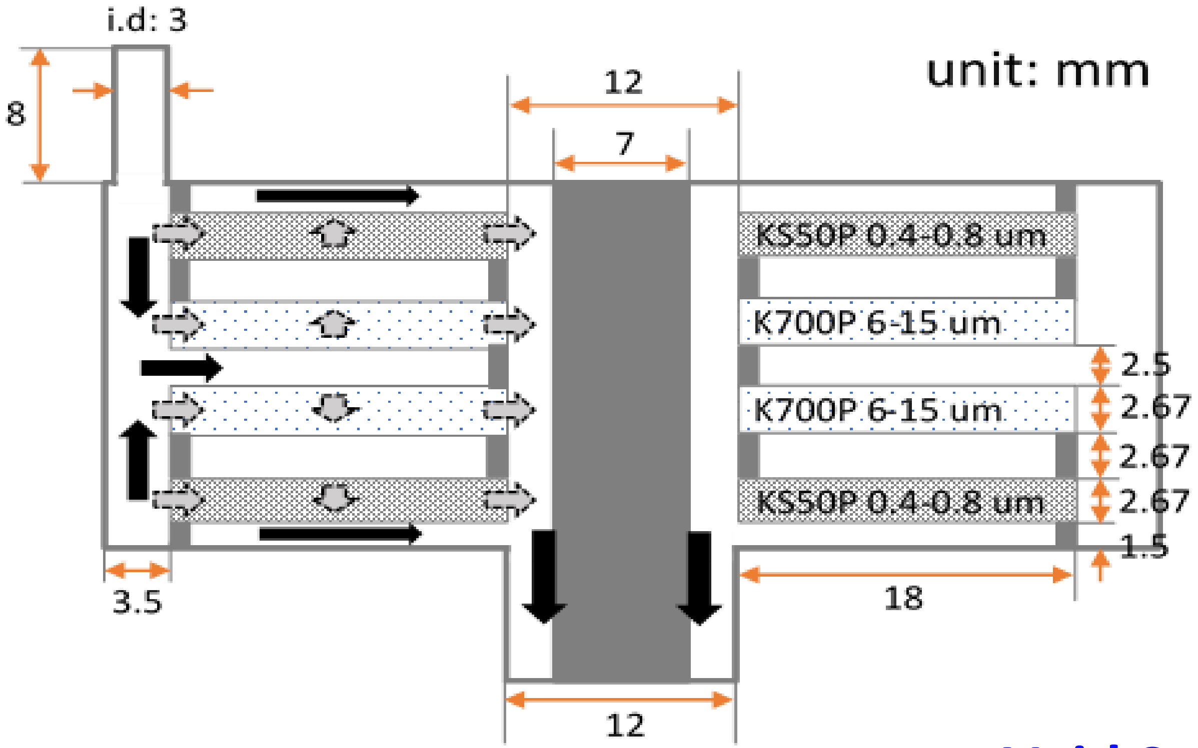
CFD Analysis

PDH4 capsule
22 cm^2 surface area



Depth Filter Media

$$\rho(\boldsymbol{u}^* \cdot \nabla \boldsymbol{u}^*) = -\nabla p + \mu \nabla^2 \boldsymbol{u}^* - \frac{\mu}{\kappa} \boldsymbol{u}^*$$

$$\nabla \cdot \boldsymbol{u}^* = 0$$

Void Space

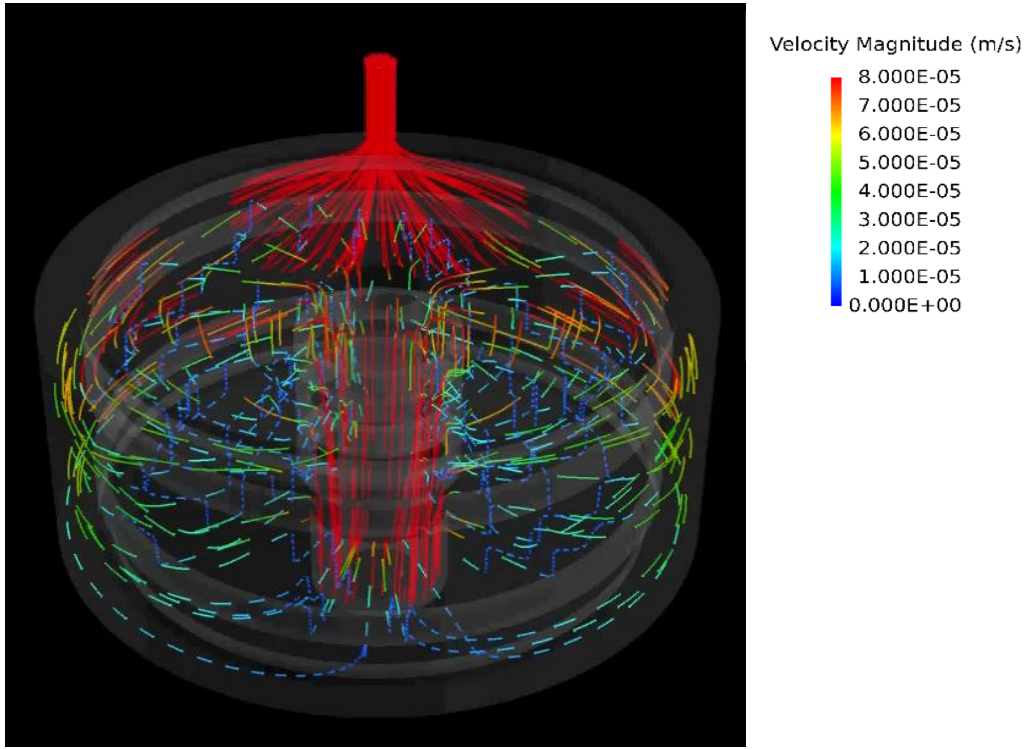
$$\rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u}) = -\nabla p + \mu \nabla^2 \boldsymbol{u}$$

$$\nabla \cdot \boldsymbol{u} = 0$$

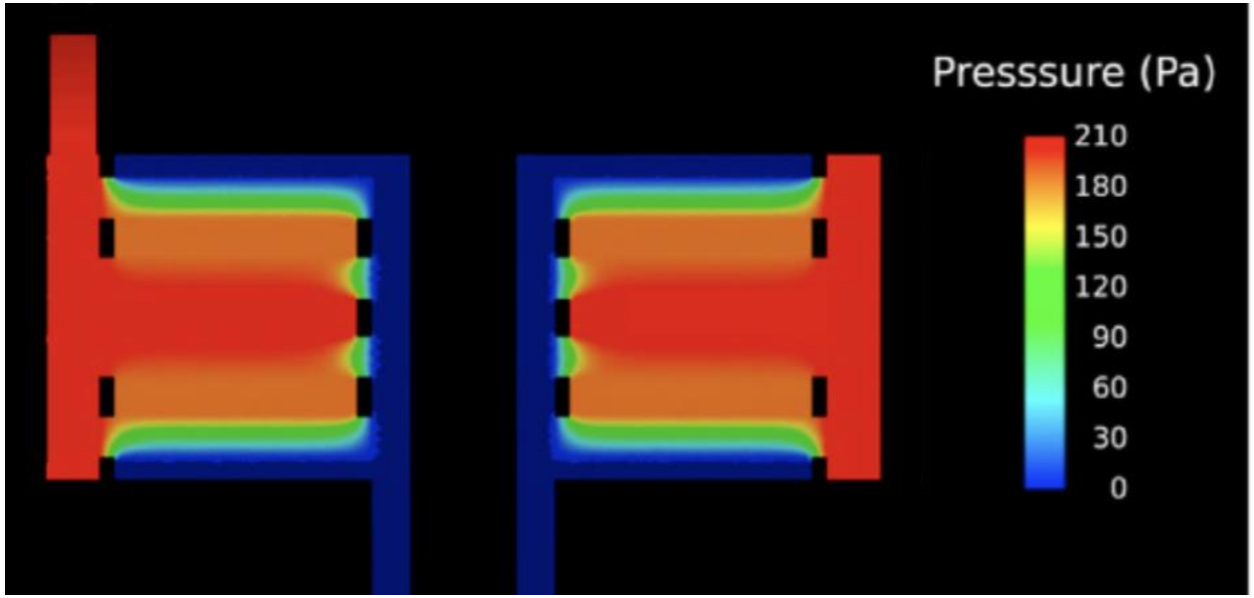
Fluid velocity and pressure profiles

CFD model suggested different flow paths in the system

Flow Streaklines

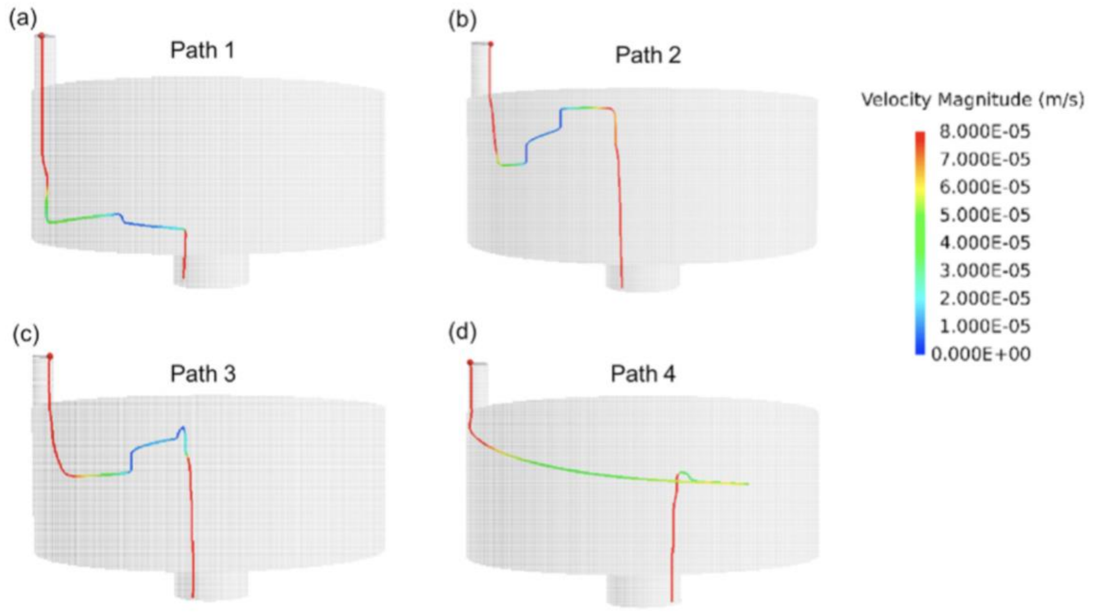


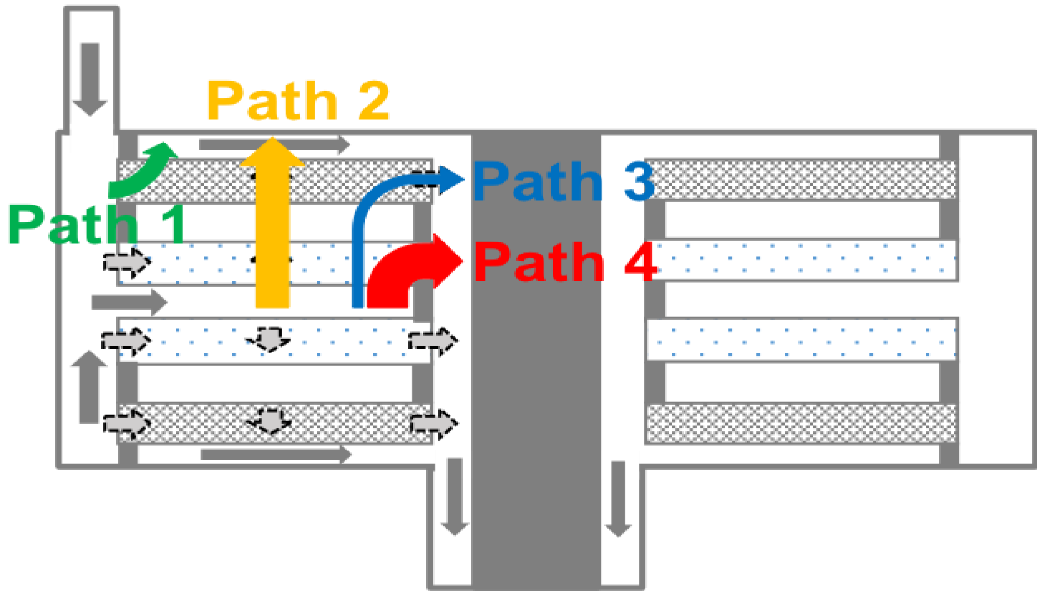
Pressure Distribution / 1.0 ml/min



- Heterogenous flow distribution, asymmetrical

Multiple Flow Paths



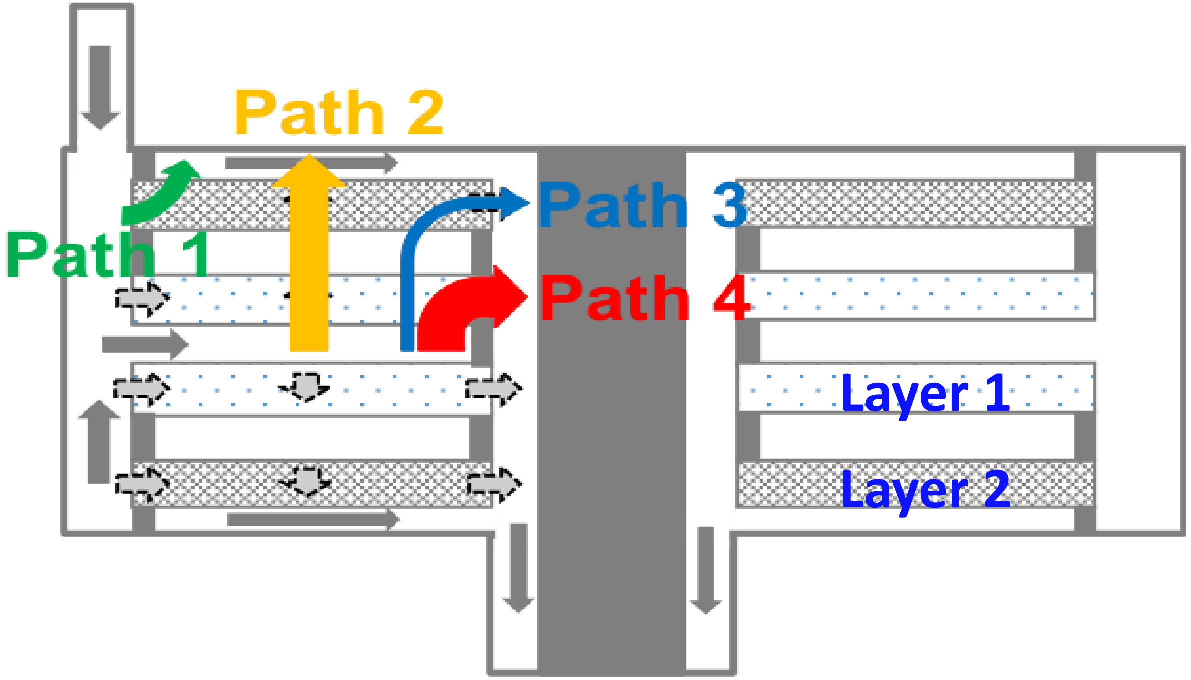


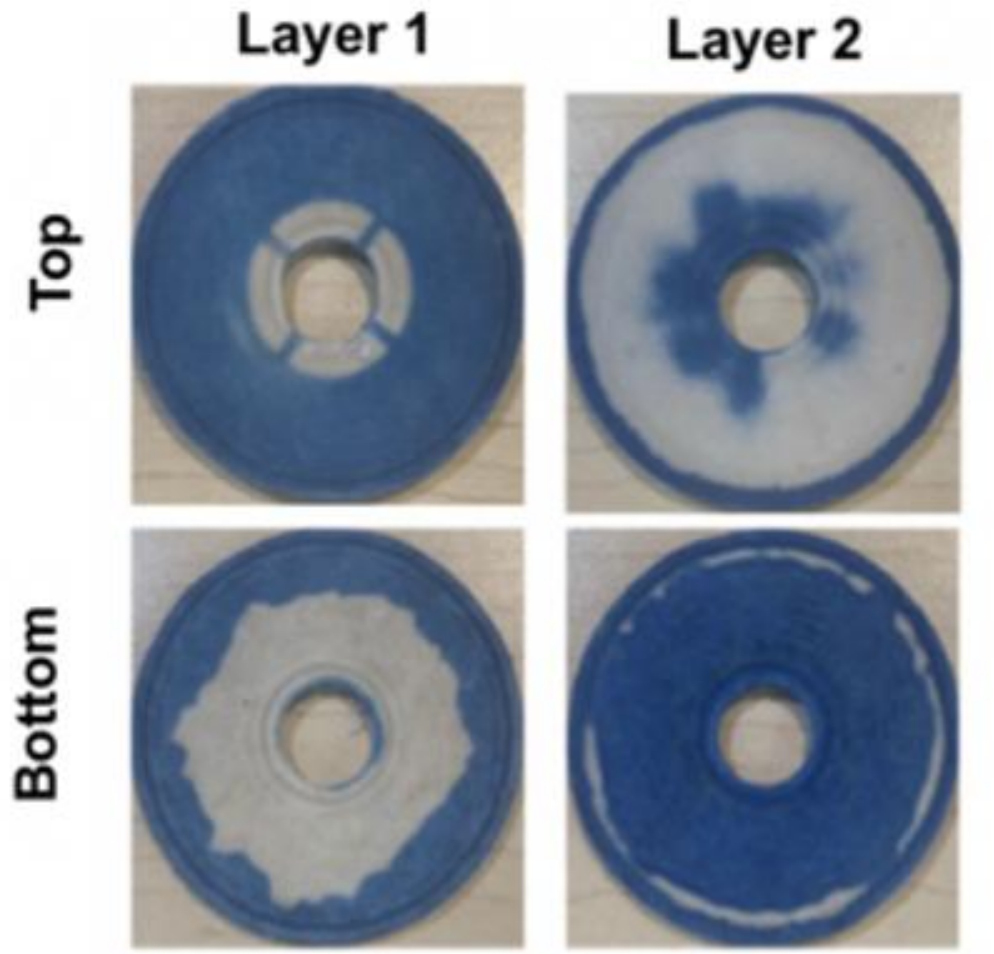
Flow fraction:

Flow Path	% of Total Flow
Path 1	10.5
Path 2	36.0
Path 3	2.3
Path 4	51.2

Dye Binding Studies

- Cibacron Blue dye filtered through capsule
- Capsule cut-open to examine filter layers



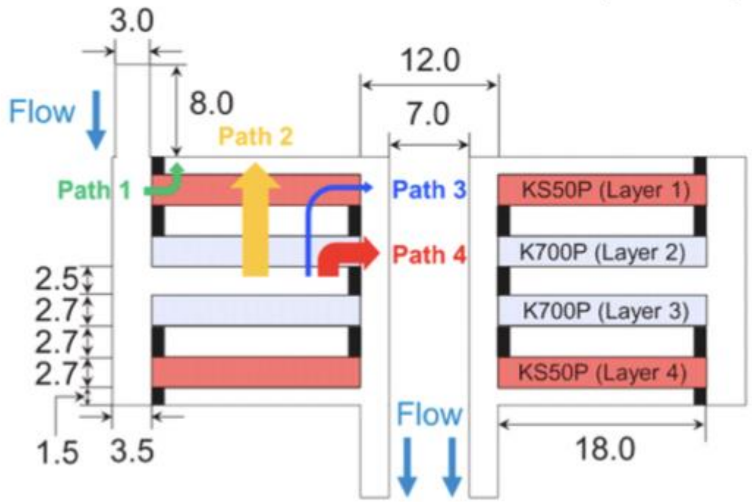


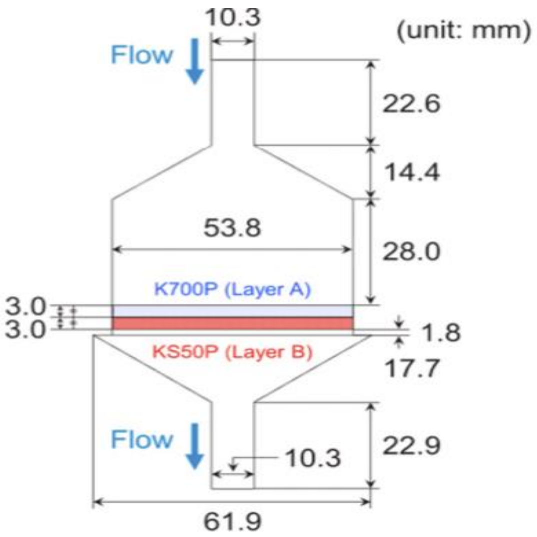
Extend CFD model to predict protein breakthrough curves

Effect of module geometry on protein purification

Supracap^{TM} 50 module

Stainless-steel module

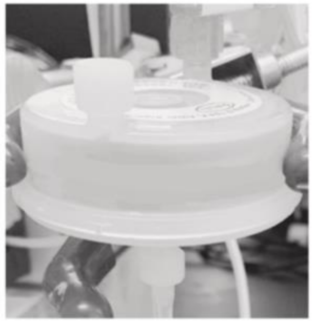


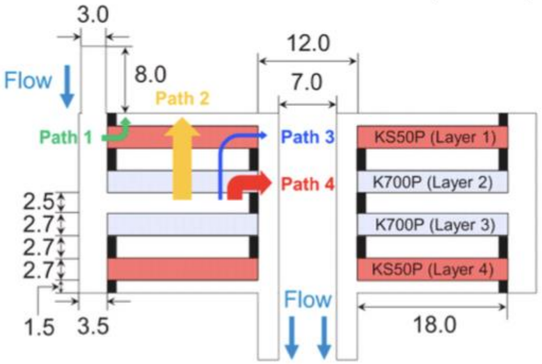


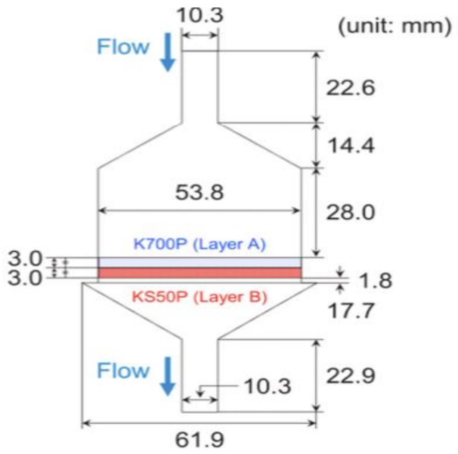
Extend CFD model to predict protein breakthrough curves

Effect of module geometry on protein purification

Supracap™ 50 module

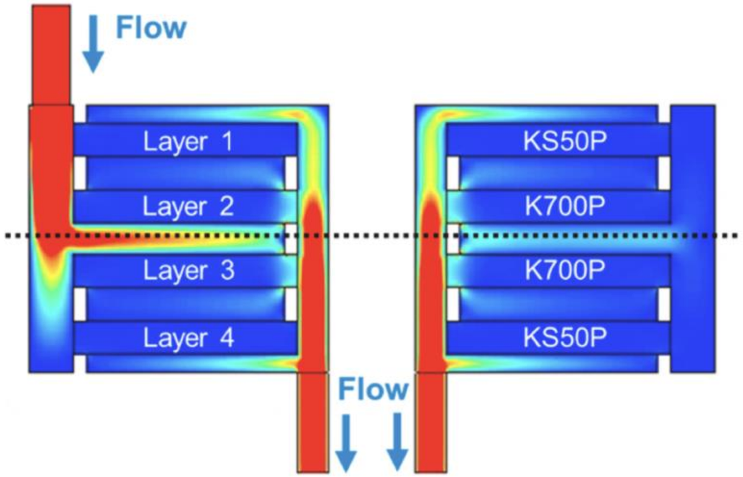


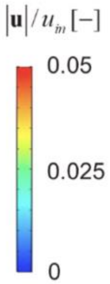


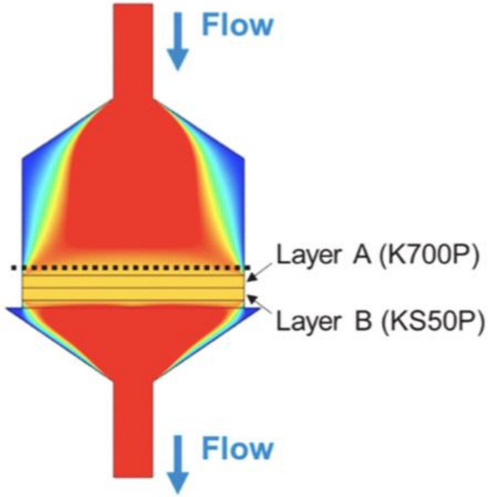


Stainless-steel module

Evaluate flow structure in both geometry





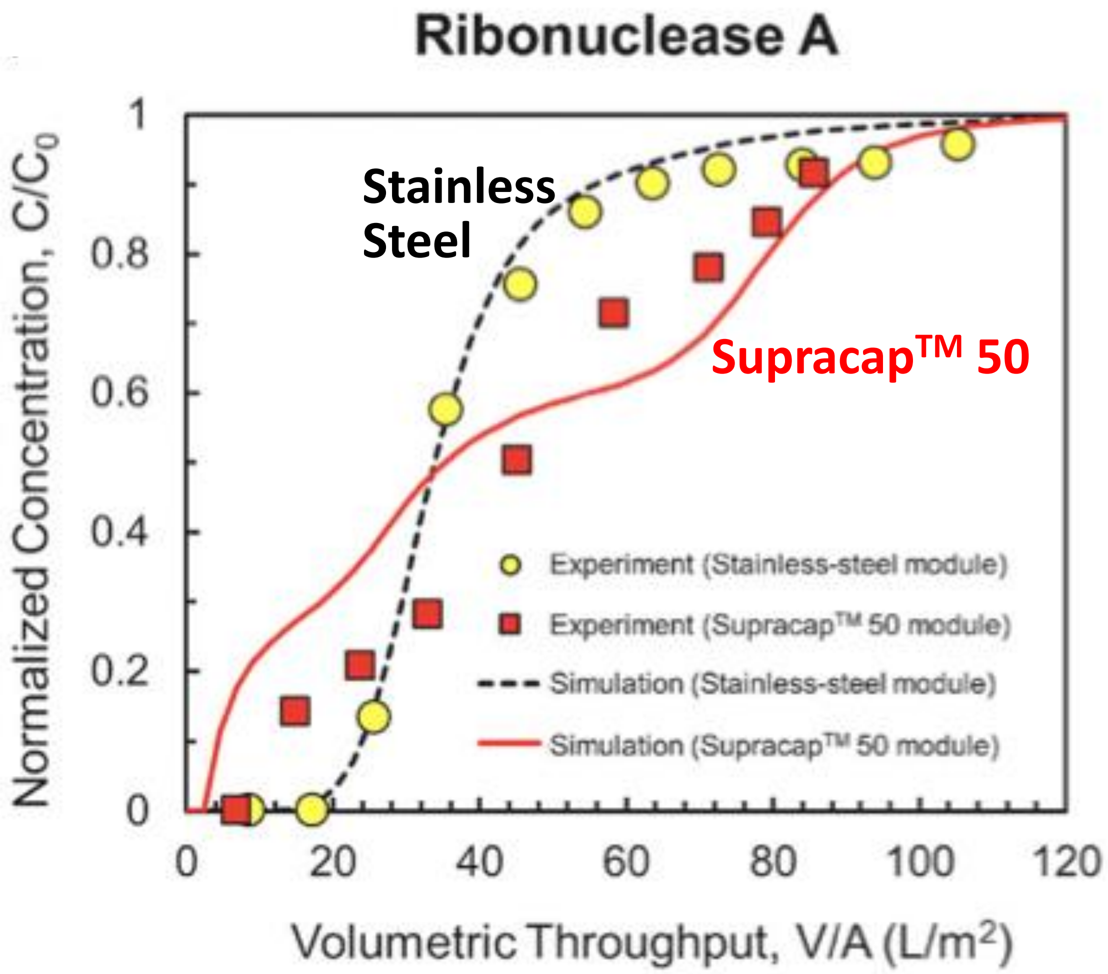


None uniform flow distribution

Uniform flow distribution

Protein Breakthrough

Dynamic binding capacity strongly influenced by flow distribution



- Langmuir binding kinetics used to account for protein binding

$$\frac{C_{eq}}{q_{eq}} = \left(\frac{1}{q_m}\right) C_{eq} + \left(\frac{1}{q_m}\right)\left(\frac{k_d}{k_a}\right)$$

- Stainless-steel module → sharper breakthrough curve with longer lag-time for initial breakthrough

Depth Filter Scale-Up

Pall Stax™

Area: 0.25 m^2

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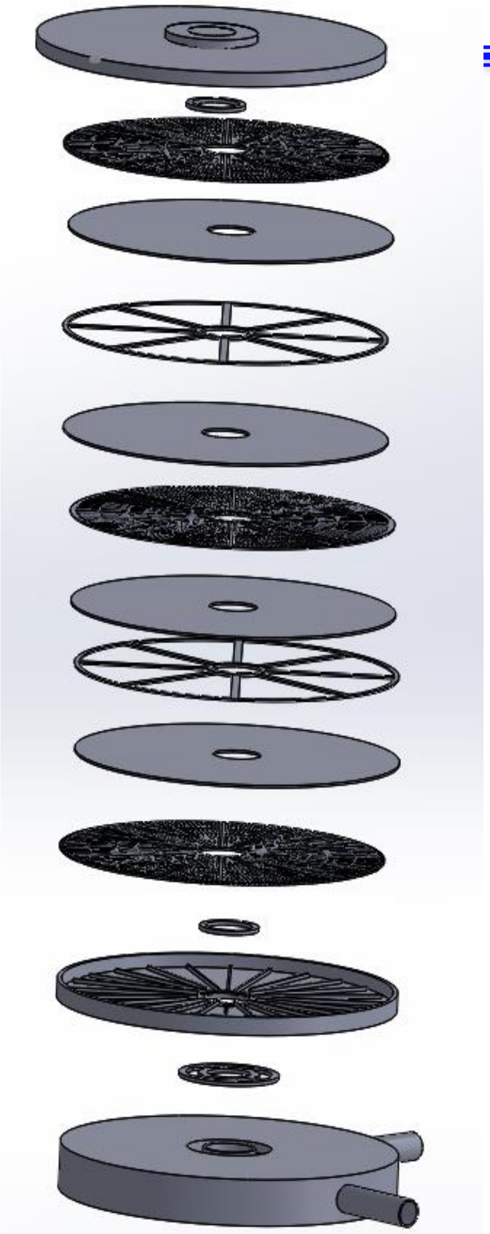
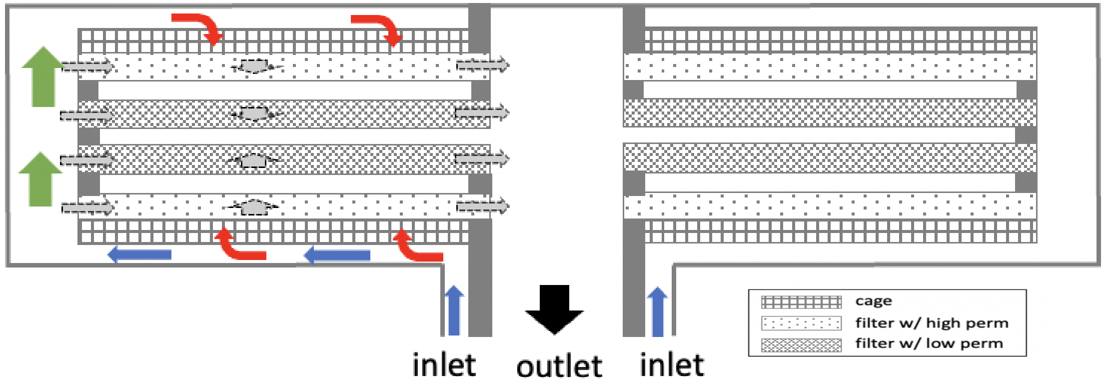


Developing CFD model for large-scale depth filter

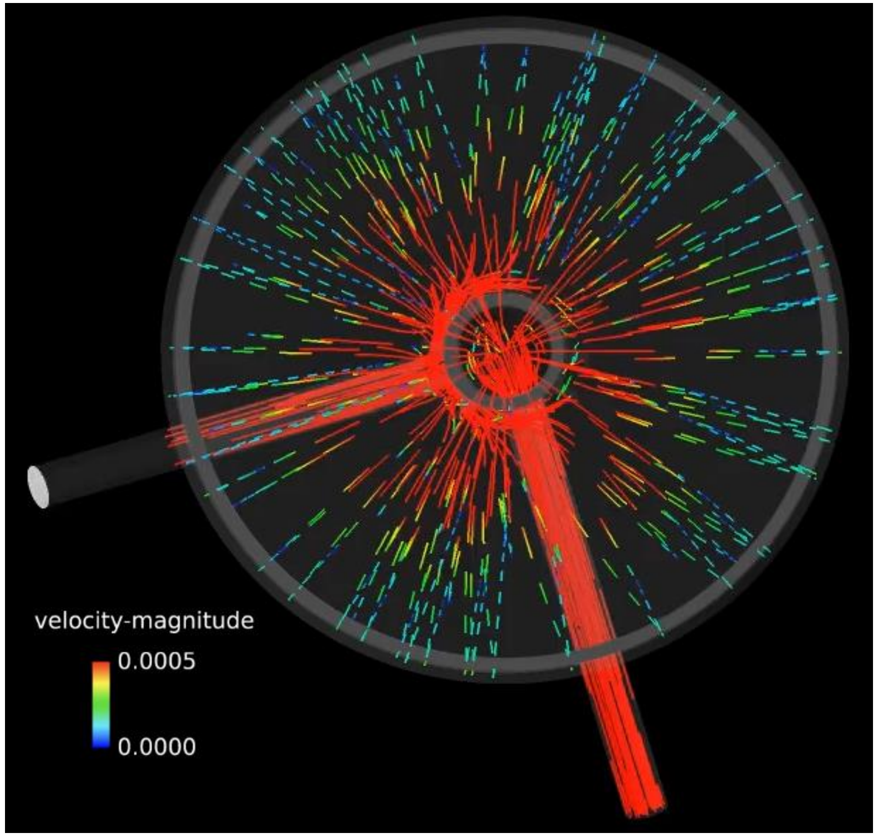
CFD model developed for flow and pressure profiles within the large-scale capsule

Stax module configuration



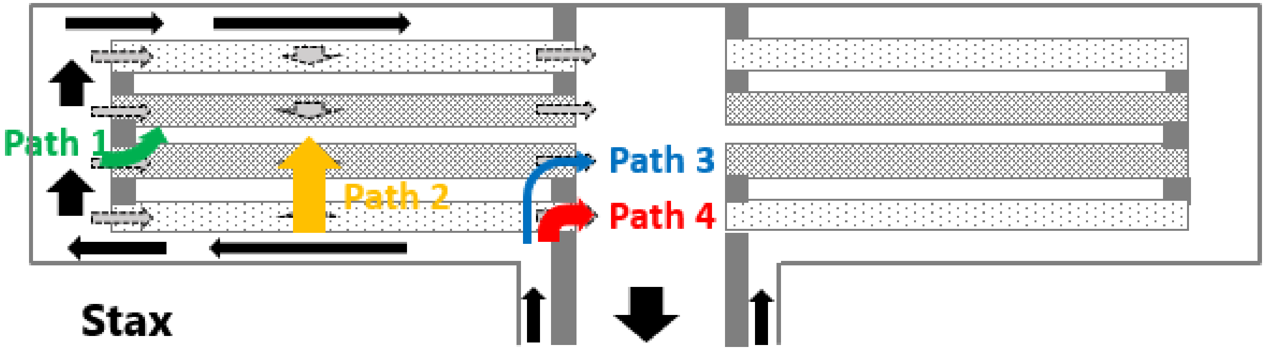
Inverted flow order

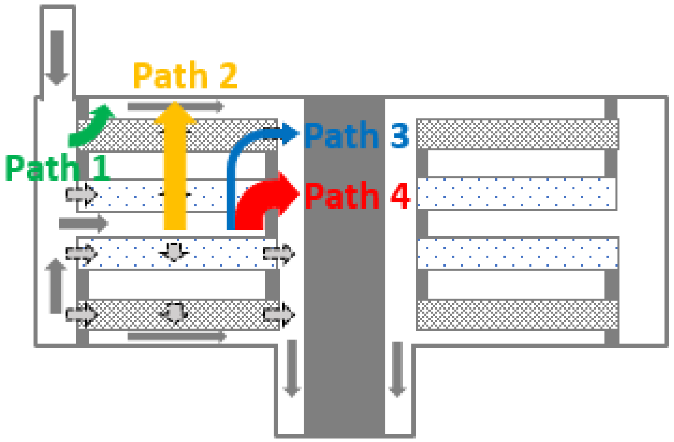
Streaklines in Stax module



Scale-up Effect: Flow Paths

Large-scale has much more uniform flow distribution

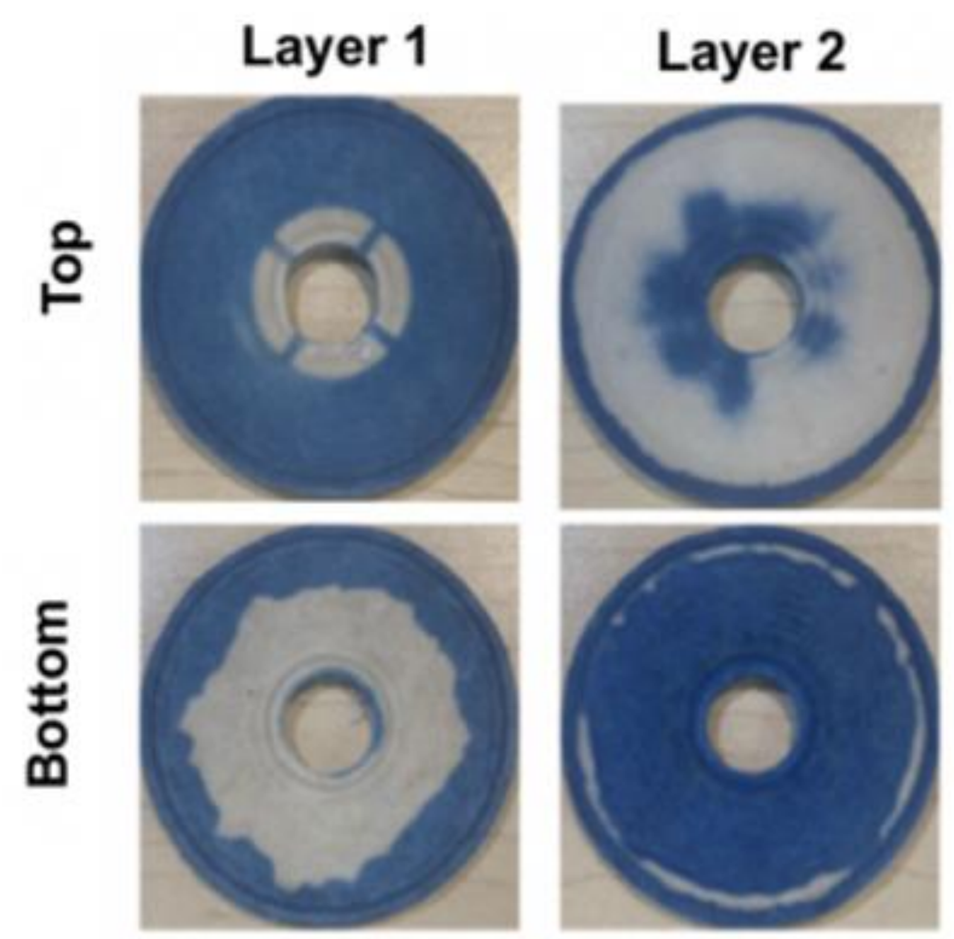




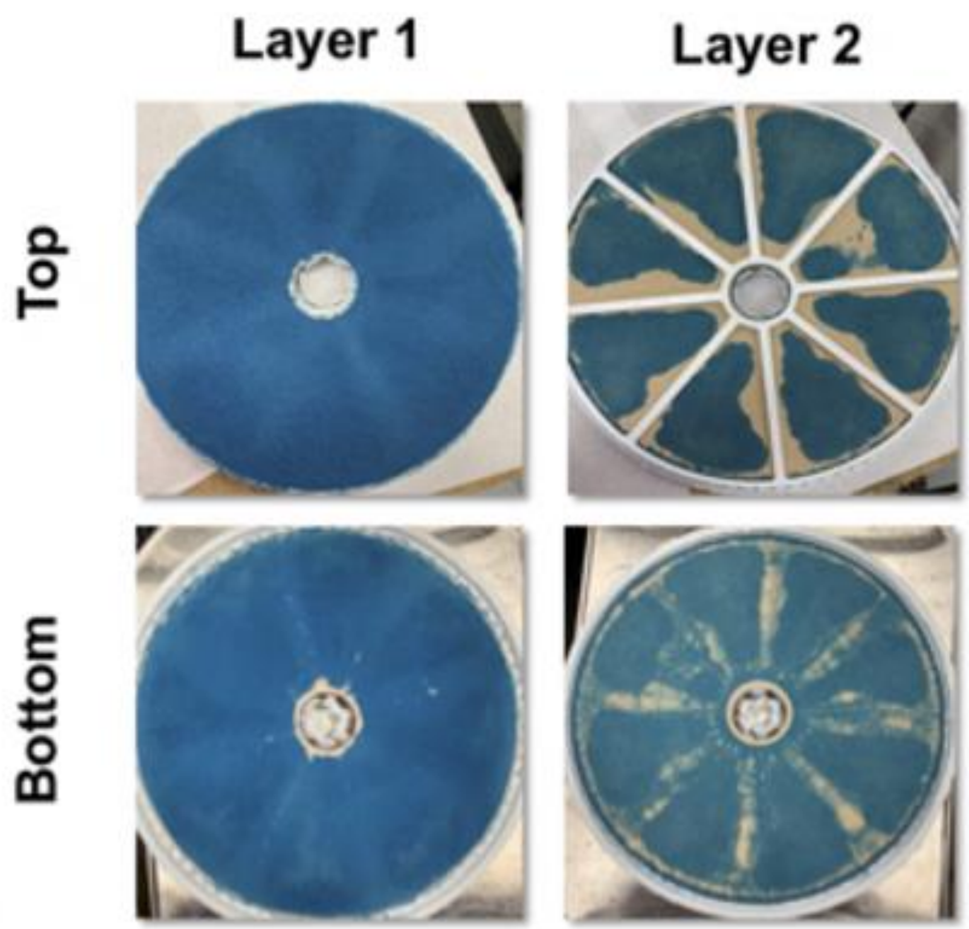
	Flow Fractions	
	Stax	SupraCap
Path 1	0.4	10.5
Path 2	97.2	36.0
Path 3	0.01	2.3
Path 4	2.4	51.2

- Path 2 is desired flow path in both modules
- Stax module shows very little bypass (<2%) due to the small relative surface area for edge of Paths 3 and 4

Scale-Up Effects: Dye Binding Studies

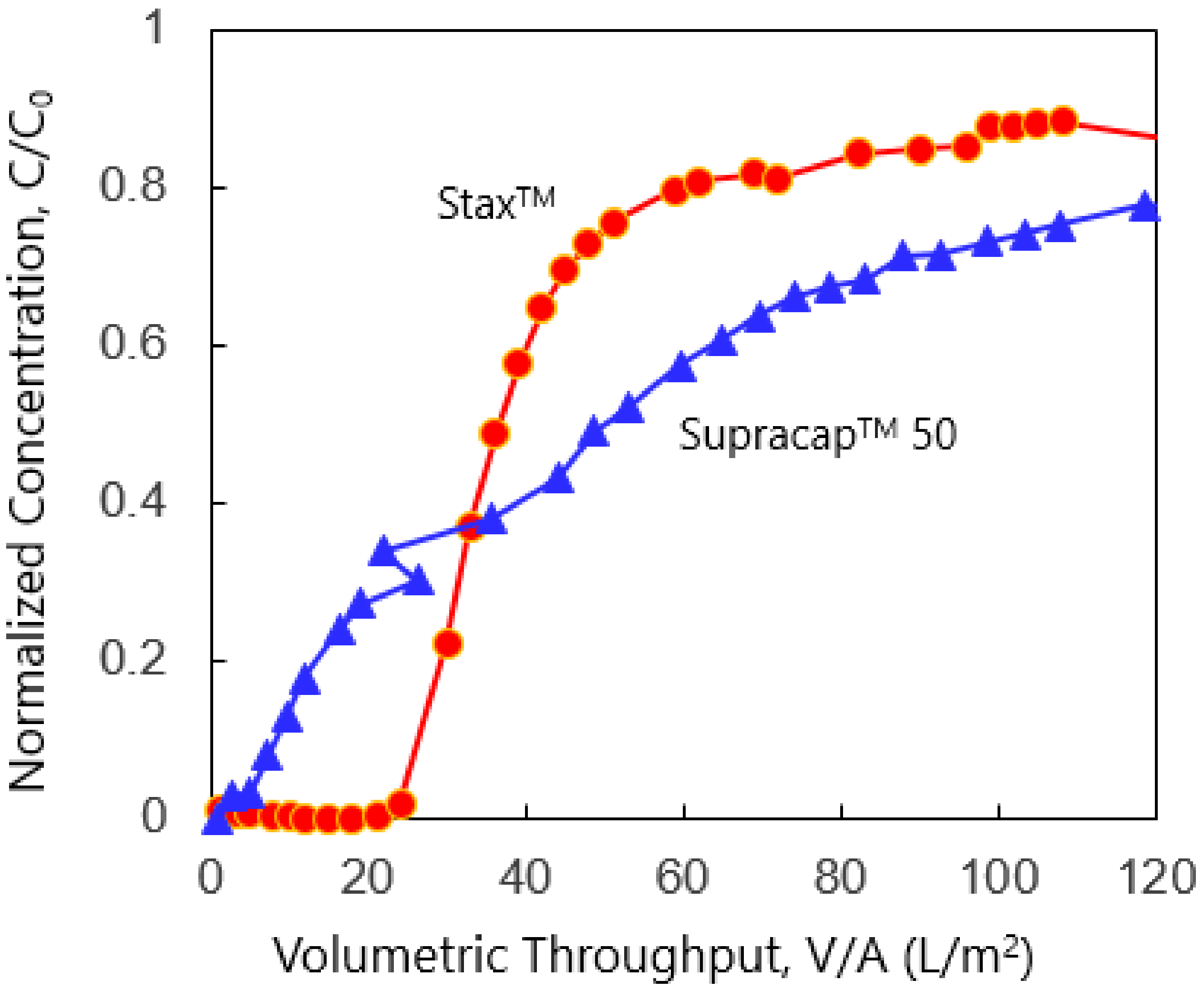


Supracap (small-scale)



Stax module (large-scale)

Scale-Up Effects: Residence Time Distribution



- Residence time distribution evaluated using non-interacting tracer
- Supracap™ 50 module shows faster initial breakthrough due to bypass flow
- Stax™ module shows much sharper breakthrough curve due to more uniform flow distribution

Cell culture fluid (CCF) clarification

Small-Scale – PendoTech

Large-Scale – PendoTech

- CHO cell culture: 13 x 10^6 cells / mL, 60% viability
- Constant flux filtration: 0.6 and 1.2 L/m^2/min

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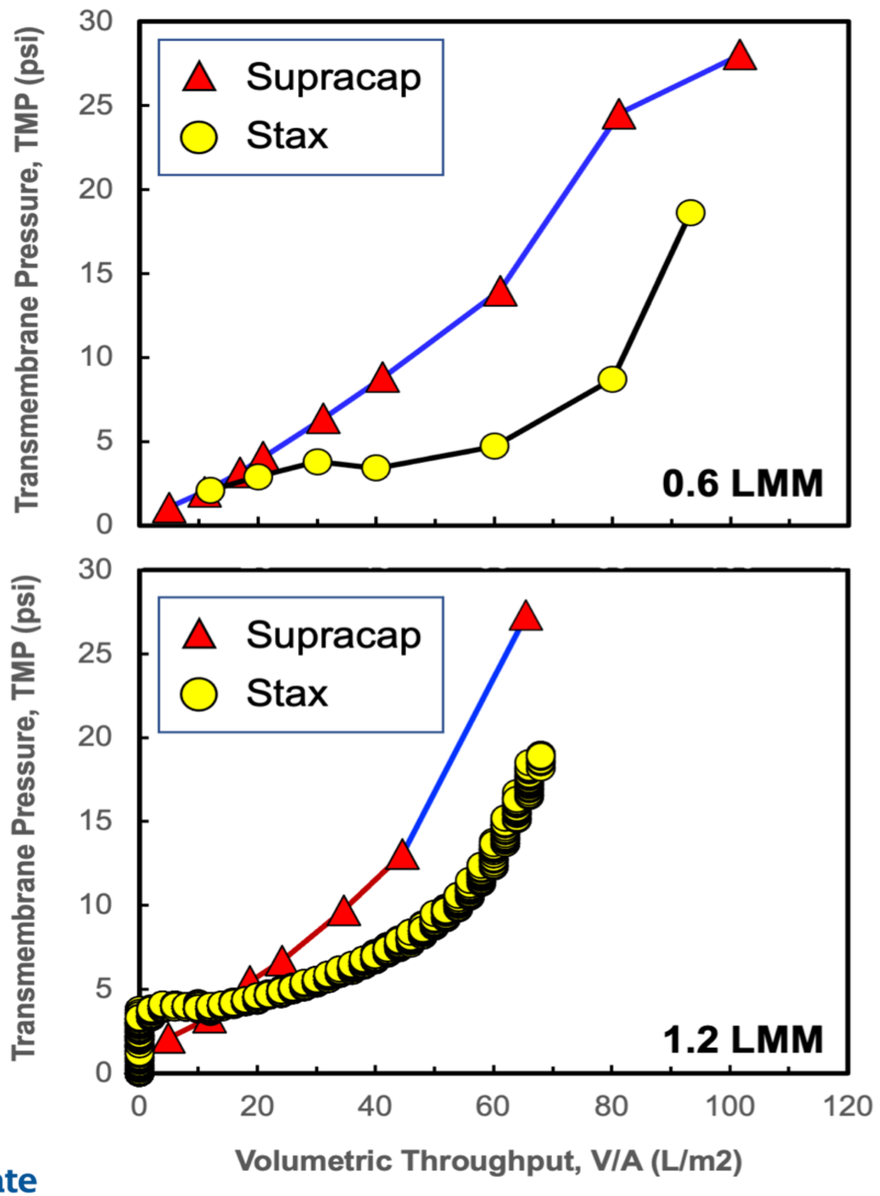
TMP profiles



- Pressure increases faster in the small-scale Supracap module due to non-uniformities in flow distribution
- Lower capacity at the higher filtrate flux → likely due to compressibility of deposited cells

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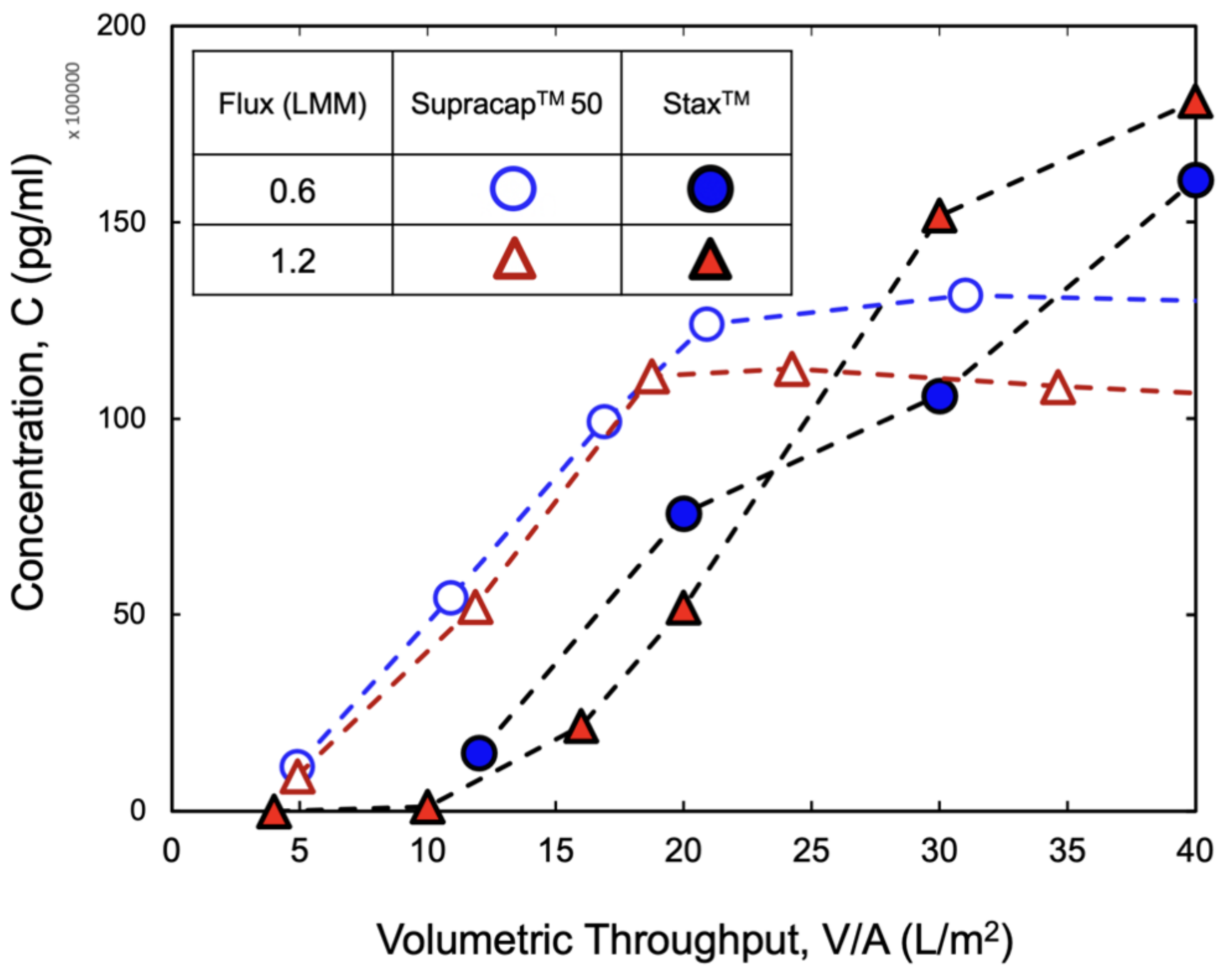
DNA removal / breakthrough



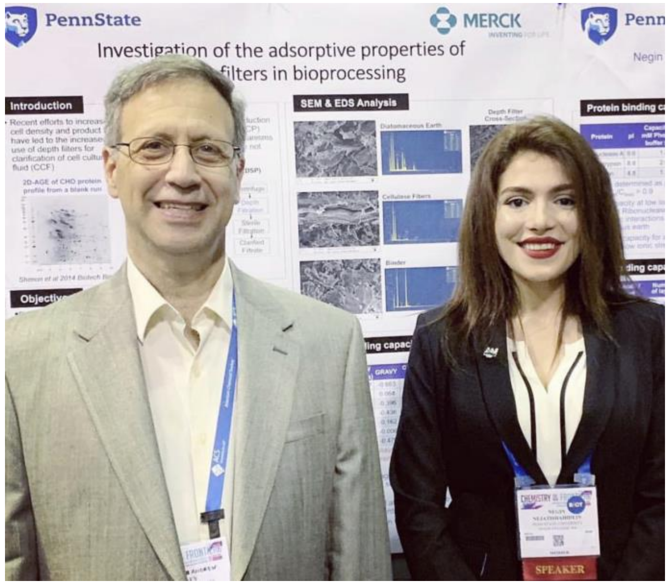
- ➢ More rapid DNA breakthrough in Supracap module due to flow bypass
- ➢ Results validated that the flow path is more uniform in the large-scale module

Conclusions

- CFD model developed to describe flow profiles in both scale-down and pilot-scale depth filtration modules:
  - ✓ Significant bypass flow in small-scale module
  - ✓ Predictions validated using dye-binding and RTD studies
- Protein / DNA breakthrough curves evaluated by combining CFD analysis with Langmuir binding isotherm
  - ✓ Model predicts diffuse breakthrough in small-scale module
- Implications for scale-up
  - ✓ More uniform flow in pilot-scale module leads to greater capacity (based on TMP) and sharper breakthrough curves
  - ✓ General approach can lead to more accurate scale-up with reduced filter requirements

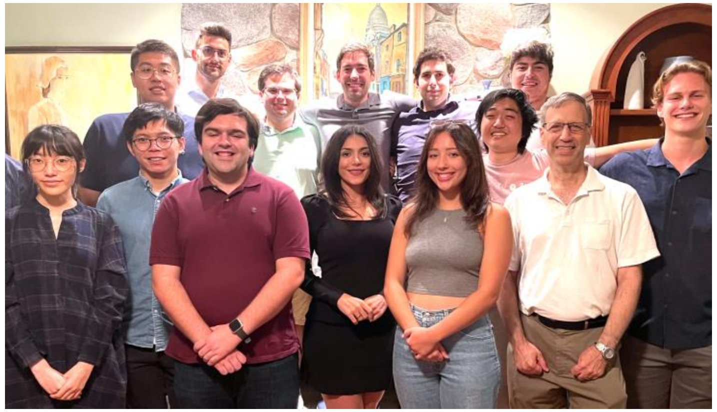
Acknowledgments

Prof. Andrew Zydney



Prof. Ali Borhan

Dr. Zydney's group



Merck colleagues

- Funding and assistance with the in-site experiments

MERCK
INVENTING FOR LIFE

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